



HERBAL AI

A Digital Repository of Philippine Herbal Medicine
with AI-Assisted Preparation Guides

Capstone Defense Presentation

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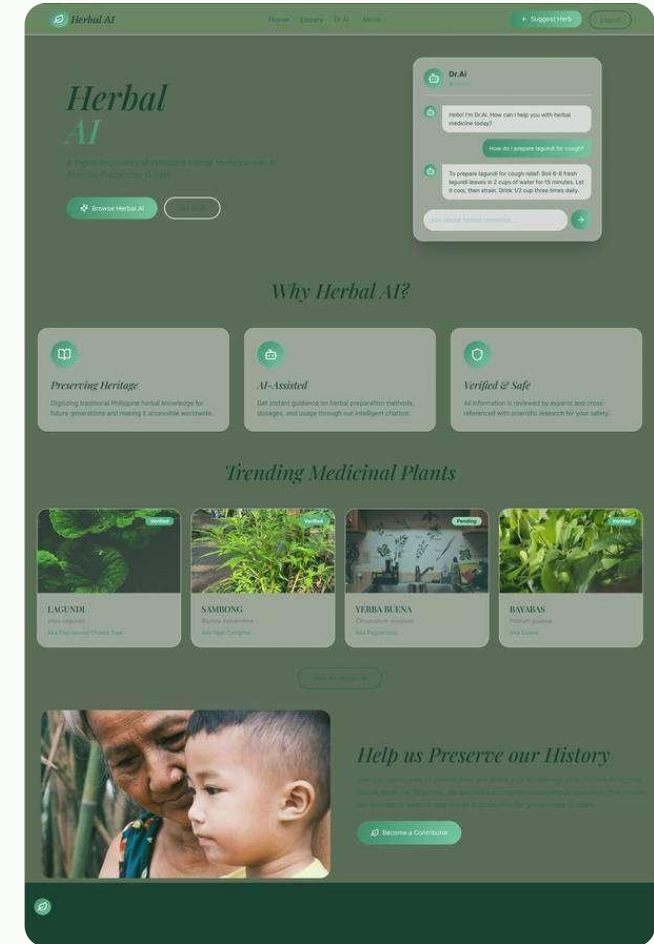
Bachelor of Science in Information Technology

The problem Herbal AI addresses

02

Traditional knowledge is useful, but access is fragmented and safety context is inconsistent.

- Herb information is scattered across oral tradition, social posts and disconnected references.
- People may confuse similar plants or follow preparation advice without dosage and warning context.
- Community knowledge needs a review process before it becomes public.



Project response

One searchable repository with moderated content and grounded AI explanations.

Objectives and system scope

03

Verified herb library

Search approved Philippine medicinal plants by name, category and medicinal use.

Dr. AI

Retrieve approved repository facts, then explain them clearly with safety limits.

Community contribution

Accept herb suggestions and publish them only after administrator review.

Account and collaboration

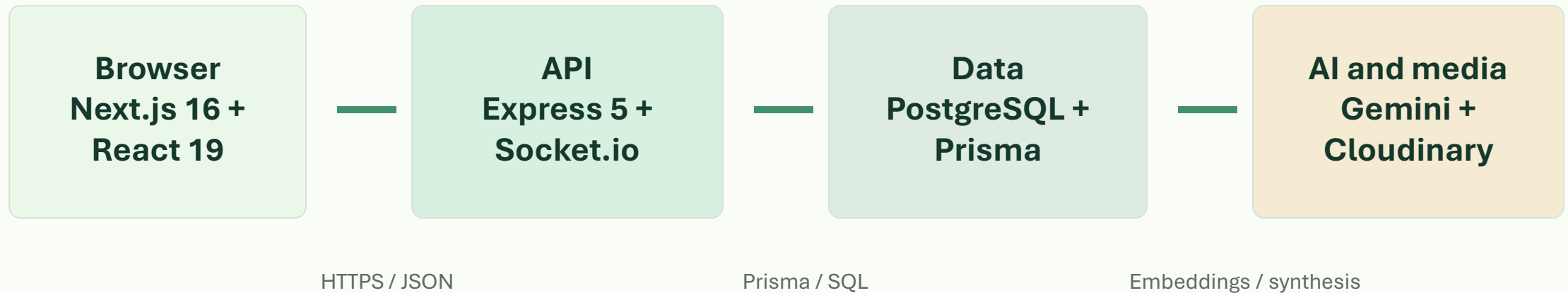
Support secure sessions, profiles, discussions, comments and private messages.

Scope boundary

The system provides traditional-knowledge guidance. It does not diagnose, prescribe or replace professional medical care.

System architecture

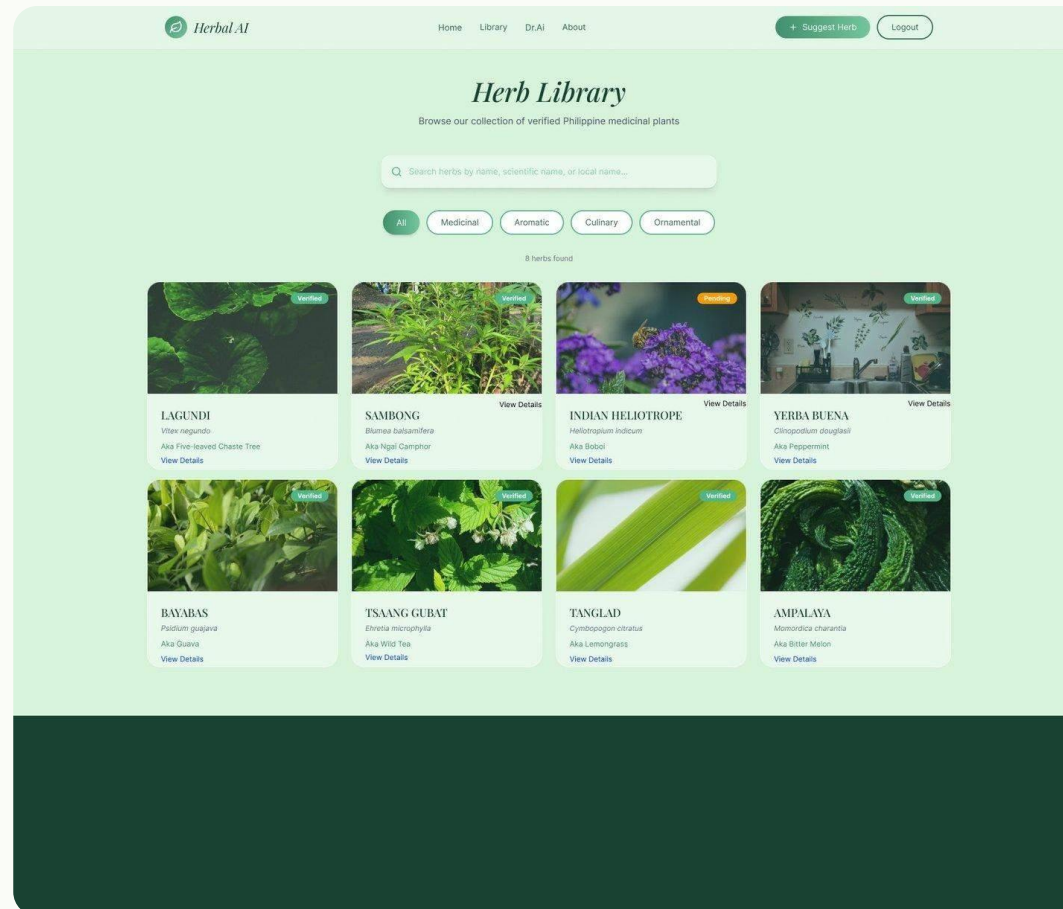
04



- The API owns validation, authorization and external service access.
- Socket.io delivers authenticated messages and notifications.
- Secrets stay in ignored backend environment files.

Verified medicinal-plant library

05



Repository functions

- Approved entries only in the public catalog
- Search by local, Cebuano or scientific name
- Category filters and paginated results
- Preparation, dosage and warning details
- Verified status and meaningful image text

10

DOH validated plants seeded in the catalog

Dr. AI uses retrieval before explanation

06

1

Retrieve

Find relevant approved herb and knowledge-base records.

2

Constrain

Reject unrelated sources and retain dosage, warnings and scope.

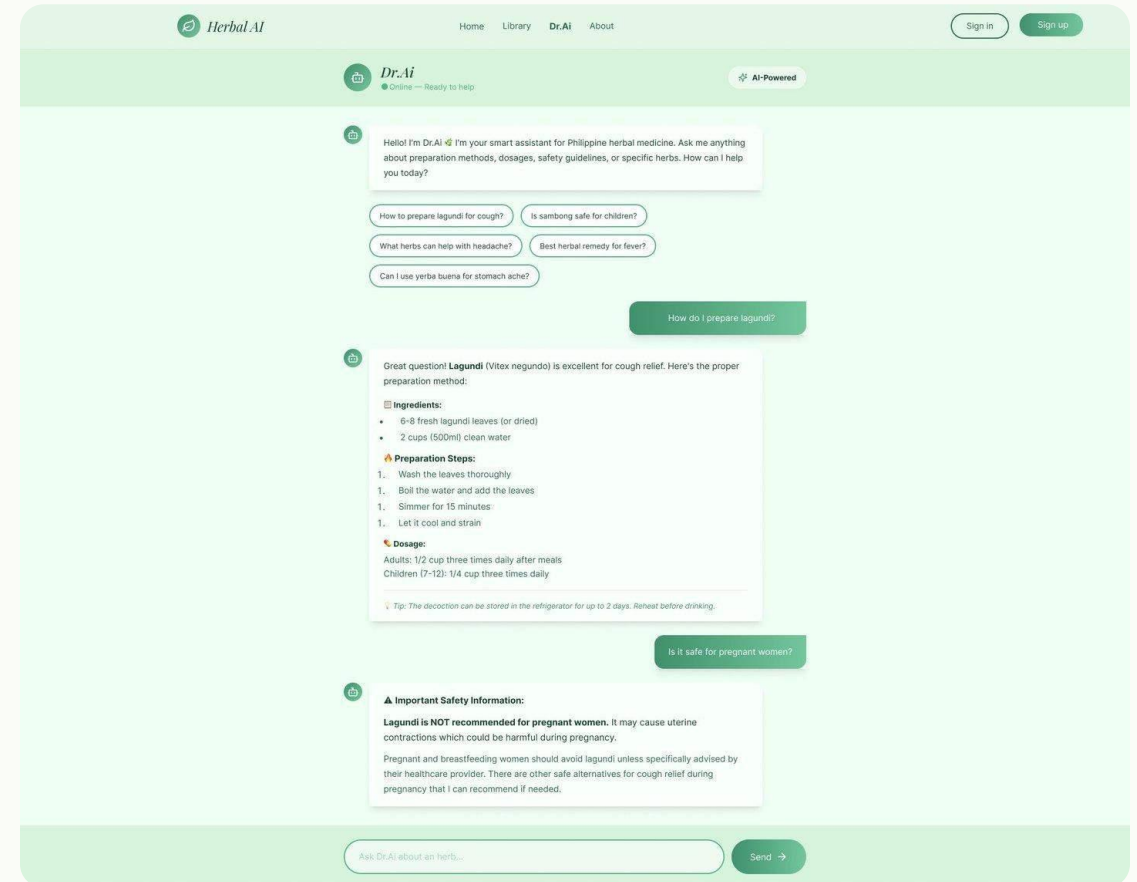
3

Explain

Rewrite the short source text into a clear answer with attribution.

Safety rule

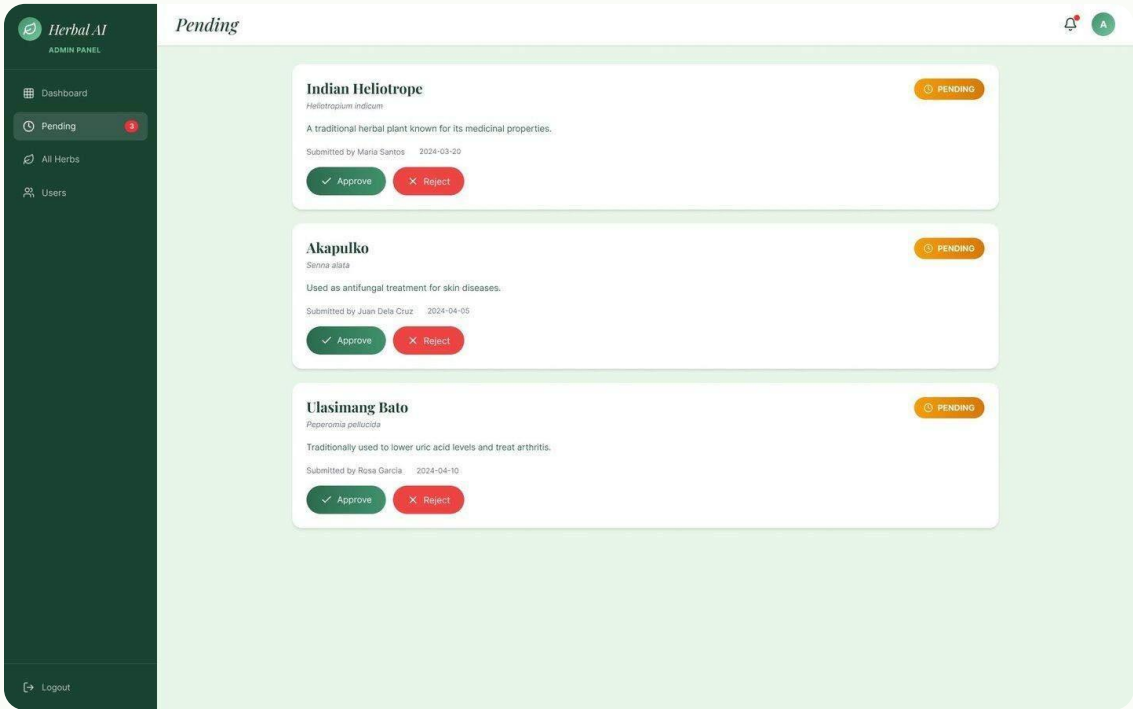
No relevant approved source means no invented herb dosage.



Contributor and administrator workflow

07

- 1 Contributor submits
- 2 Pending review
- 3 Admin approves or rejects
- 4 Catalog and notification update



Verified browser lifecycle

Create, publish, edit, RBAC denial, audit and delete all passed with temporary-data cleanup.

Roles and collaboration features

08

Guest

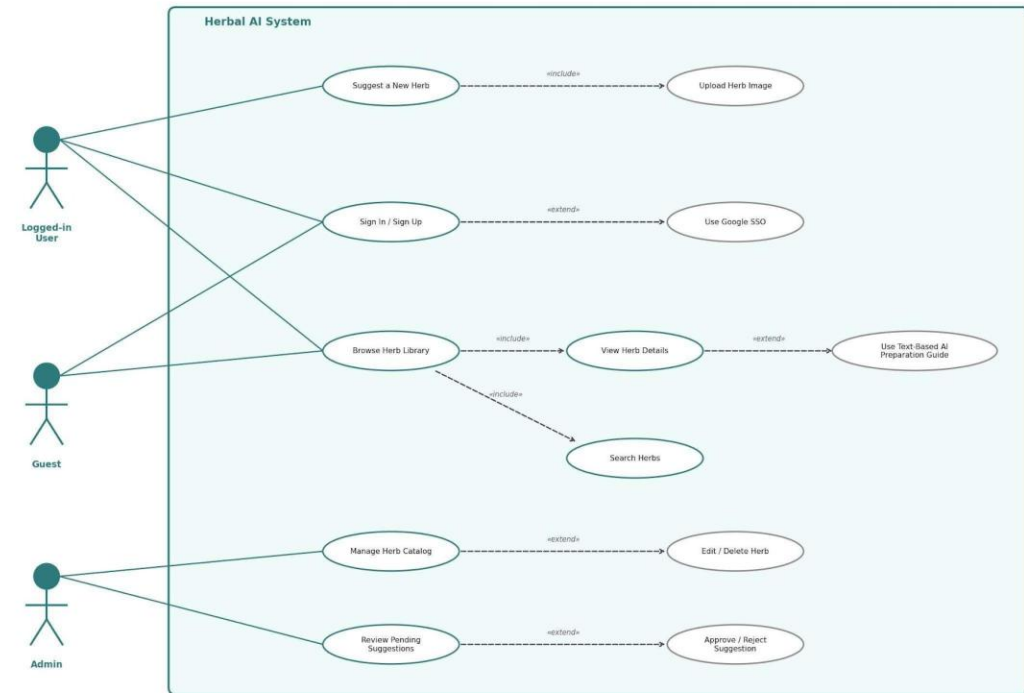
Browse verified herbs, view details and open sign-in.

Contributor

Use Dr. AI, suggest herbs, join discussions, comment and message.

Administrator

Moderate submissions, manage users, herbs, knowledge and audit history.



Real-time evidence

Two authenticated users exchanged messages and received Socket.io delivery without reloading.

Security and trust controls

09

Authentication

Short access tokens, rotating refresh tokens and logout revocation

Authorization

Server-side role checks protect every administrative route

Input safety

Validation, parameterized database operations and upload signature checks

Secret handling

Backend-only keys, ignored environment files and sanitized production errors

AI safety

Approved-source retrieval, no-source refusal and medical disclaimer

Account recovery

Single-use verification and reset links with expiry and concurrency protection

Actual Gmail verification and password-reset links passed with isolated test-account cleanup.

Verification baseline

83 / 83

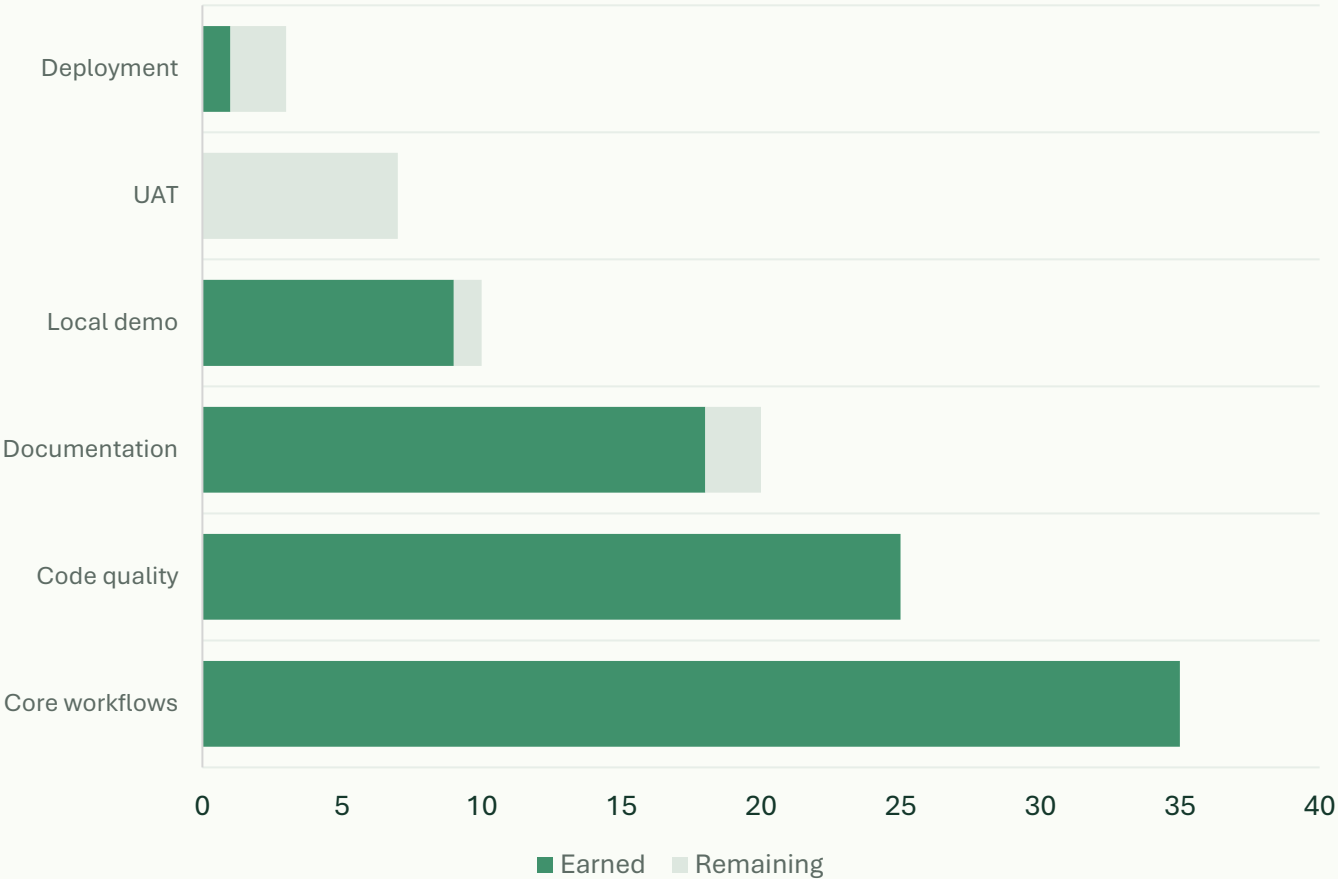
backend tests passed across 14 files

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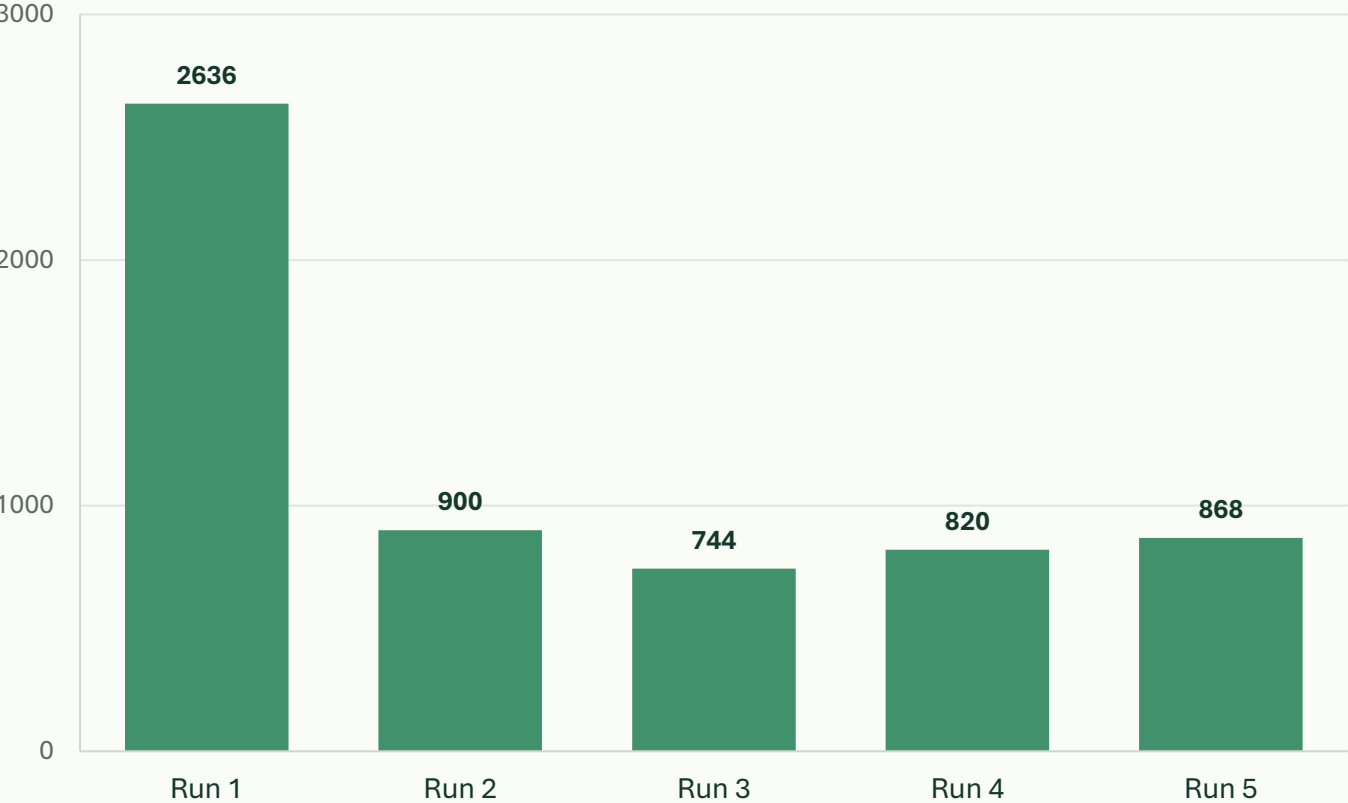
functional API checks passed

26 / 26

emulated accessibility width scans passed



Local performance improved, staging remains open



Local PR-001 result

2,636 ms

five-run local p95 / maximum
Target: 3,000 ms

Post-login profile

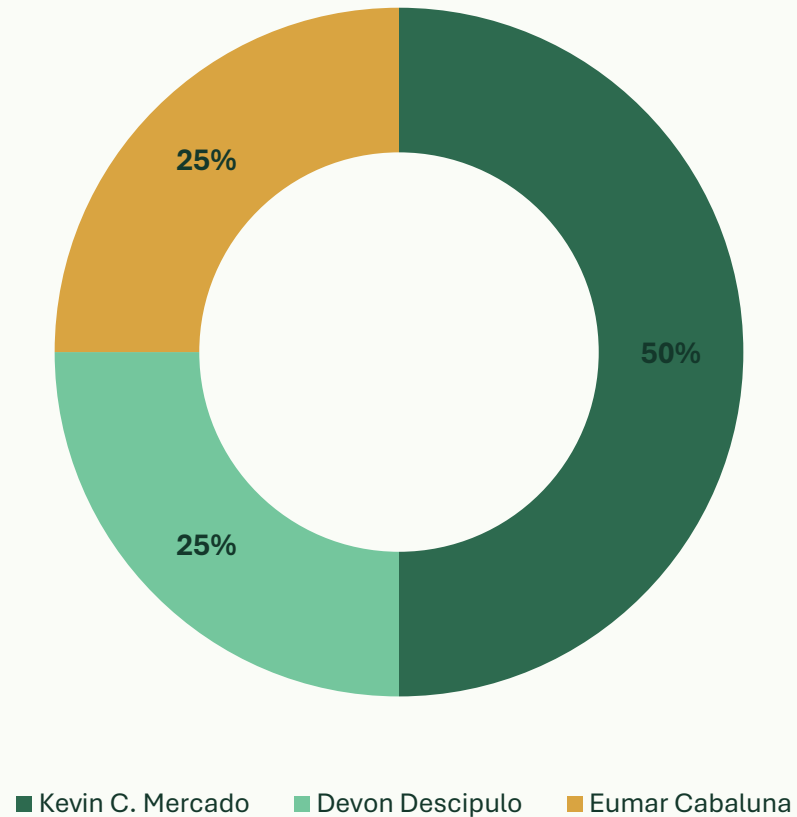
37.5 ms

first /auth/me end-to-end
9.1 ms inside Express

PR-004 and PR-005 still require database-proximate staging.

Team contribution allocation

12



Kevin C. Mercado

50%

Project Manager and Backend Developer
Planning, architecture, authentication, database, integration, testing and documents

Devon Descipulo

25%

Backend Developer
API, database and AI integration support

Eumar Cabaluna

25%

Frontend Developer
Responsive UI, frontend integration and usability support

Current readiness and remaining gates

13

88

out of 100

Ready for a controlled local capstone presentation

The score measures internal evidence. The adviser and panel make the academic decision.

Remaining acceptance gates

- Five real UAT participants and signed summary
- Staging performance for PR-004 and PR-005
- Production deployment, TLS and recovery evidence
- Physical-device and full WCAG review
- Adviser and panel signatures

These items remain visible in the documentation. The project does not claim 100% acceptance.

Defense demonstration flow

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- 01 Public catalog** Search Lagundi and open its verified details.
- 02 Dr. AI** Ask a preparation question and point to the source and disclaimer.
- 03 Contributor** Show suggestion, profile, community and Messenger flows.
- 04 Administrator** Review moderation, herb/knowledge management and audit history.
- 05 Evidence** Close with tests, performance limits and readiness gates.

Fallback plan

Keep screenshots ready for Google OAuth, email and Gemini if the presentation network fails.



HERBAL AI

Verified local functionality, clear evidence boundaries and a practical path to final acceptance

Questions?

Kevin C. Mercado
Project Manager and Backend Developer